

Pranitha Padala

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Profile

Detail-oriented and analytical Computer Science graduate with a strong foundation in data analysis, statistical techniques, and data visualization. Experienced in working with large datasets to uncover trends, patterns, and actionable insights using tools like SQL, Python, Power BI, and Tableau. Adept in data cleaning, data interpretation, and presenting findings to support data-driven decision making. Passionate about contributing to business success through meaningful analytics and collaborative teamwork.

Technical Skills

- **Languages:** Python, SQL
- **Data Analytics:** Data Cleaning, Exploratory Data Analysis (EDA), Trend Analysis, Reporting
- **Visualization Tools:** Tableau, Power BI, Excel, Matplotlib
- **Statistical Techniques:** Hypothesis Testing, Regression Analysis, Classification Models
- **Tools & Libraries:** Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, Jupyter
- **Other Tools:** Git, GitHub
- **Soft Skills:** Strong Communication, Team Collaboration, Attention to Detail, Problem Solving

Professional Experience

Smart Bridge

Jul 2020 – Sep 2020

Machine Learning Intern

- Developed a predictive model using Logistic Regression to analyze and reduce employee attrition
- Conducted data preprocessing, feature selection, and importance analysis to identify key factors influencing turnover
- Trained and evaluated the model using scikit-learn, gaining hands-on experience in applied machine learning

Verzeo

Sep 2021 – Nov 2021

Front-End Development Intern

- Developed a dynamic Single Page Application (SPA) using Angular and TypeScript to manage student, faculty, and administrative workflows
- Implemented features such as course registration, grade management, and catalog browsing with Angular services, routing, and component-based architecture
- Designed a responsive UI using Angular Material, ensuring a seamless user experience across devices

Projects

Warfarin Dose Prediction Using Machine Learning

Python, scikit-learn, Gradio, Ridge & Logistic Regression

- Built an end-to-end ML pipeline to predict personalized warfarin dosage using demographic, clinical, and genetic data
- Trained classification and regression models; logistic regression achieved 84% AUC, and Ridge regression outperformed others for continuous dose prediction

- Deployed an interactive Gradio web app allowing real-time patient data input and visual dose recommendations

Fine-Tuning EfficientNet for Medical Image Classification

Python, TensorFlow

- Developed a pneumonia detection model (98% accuracy) with transfer learning, optimizing with data augmentation & batch normalization
- Optimized model performance through hyperparameter tuning and cross-validation

PCOS Prediction Using Machine Learning

Python, scikit-learn, Random Forest, KNN, Logistic Regression

- Developed a PCOS prediction model (83.4% accuracy) using feature selection, data preprocessing, and hyperparameter tuning
- Applied data visualization and model evaluation using confusion matrices, precision-recall curves, and decision boundaries for early diagnosis

Crop Yield Predictor & Recommender System

Python, Random Forest

- Designed a Random Forest-based predictor (90% accuracy) for real-time recommendations
- Developed a crop recommender system to enhance agricultural productivity

Education

Saint Louis University

M.S. in Computer Science — Saint Louis, US

Aug 2023 – May 2025

Sreenidhi Institute of Science and Technology

B.Tech in Computer Science — Hyderabad, India

Jun 2018 – Jun 2022

Certificates

- IBM Data Science - Coursera
- Machine Learning with Python – IBM (Coursera)
- Data Analysis with Python- Coursera
- Data Visualization with python
- Getting Started with AWS – Simplilearn